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ASSIGNMENT REPORT

**Topic: Design and Develop a DNS Topology Mapping Framework
to Visualize the Hierarchical DNS Infrastructure**

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ABSTRACT

The **Domain Name System (DNS)** is a critical **Internet infrastructure** component that translates human-readable domain names into machine-readable IP addresses. This assignment presents the design and development of a **DNS Topology Mapping Framework** that visualizes the hierarchical DNS infrastructure through systematic analysis of real-world resolution paths. DNS operates on a distributed, hierarchical database comprising **Root Name Servers**, **Top-Level Domain (TLD) Servers**, and **Authoritative Name Servers**, each performing distinct roles.

Using the **dig** utility with the **+trace** option, the complete resolution path for three Indian domains was traced: **www.iitm.ac.in**, **www.iitmandi.ac.in**, and **www.nic.in**. The framework identifies **Root**, **TLD**, and **Authoritative servers** at each level. Results reveal the delegation structure of the Indian namespace from Root through the **.in** TLD to second-level domains. Comparative analysis highlights differences in **server distribution**, **redundancy**, **TTL values**, and **resolution efficiency** across the three domains.

The framework produces topology diagrams that visually represent complete DNS infrastructure. Findings demonstrate that **nic.in** achieves the best resolution efficiency (**3 queries, 38 ms**) with **5 authoritative servers** and **DNSSEC** support, while the IIT domains use 2 authoritative servers with longer resolution paths. The study contributes insights into DNS **reliability**, **scalability**, and **fault tolerance**, offering recommendations including **DNSSEC deployment**, **Anycast implementation**, **geographic redundancy**, and **load balancing**.

Keywords: DNS, Topology Mapping, dig +trace, Root Servers, TLD, Authoritative Servers, DNSSEC, Anycast

1. INTRODUCTION

1.1 What is DNS?

The **Domain Name System (DNS)** is a hierarchical, distributed naming system serving as the foundational directory service of the Internet. Functioning as the "**phonebook of the Internet**," DNS translates human-friendly domain names such as `www.iitm.ac.in` into numerical IP addresses such as `10.3.98.2`. Designed by Paul Mockapetris in 1983 (RFC 1034/1035), DNS has evolved into a massive distributed system of millions of servers worldwide with **no single point of failure**. DNS resolves various record types: **A** (IPv4), **AAAA** (IPv6), **MX** (mail), **CNAME**, **NS**, **SOA**, **TXT**, and **SRV**.

1.2 Why DNS is Required?

DNS provides **human-readable naming**, **abstraction** (changing IPs without affecting access), **load distribution** (round-robin across multiple IPs), **mail routing** (MX records), **service discovery** (SRV records), **redundancy** (multiple authoritative servers), and **geographic distribution** (directing users to nearest servers).

1.3 Evolution of DNS

Introduced in the early 1980s to replace the centrally maintained HOSTS.TXT file, DNS was defined in RFC 882/883 (1983), superseded by RFC 1034/1035 (1987). **ICANN** was established in 1998 for DNS governance. The 2000s brought **DNSSEC** (RFC 4033-4035) to address cache poisoning and spoofing. Modern DNS adds **DNS-over-HTTPS (DoH)**, **DNS-over-TLS (DoT)**, and new gTLDs.

1.4 Hierarchical DNS Architecture

DNS follows a strict tree structure. At the apex is the **Root Zone** (13 logical Root Server clusters, `a.root-servers.net` through `m.root-servers.net`, served by hundreds of Anycast instances). The Root Zone delegates to **TLD Servers** (`.com`, `.org`, `.in`). Below are **Second-Level Domains** (`google.com`, `iitm.ac.in`), which delegate to **Authoritative Name Servers** holding actual DNS records. This hierarchy scales to millions of domains.

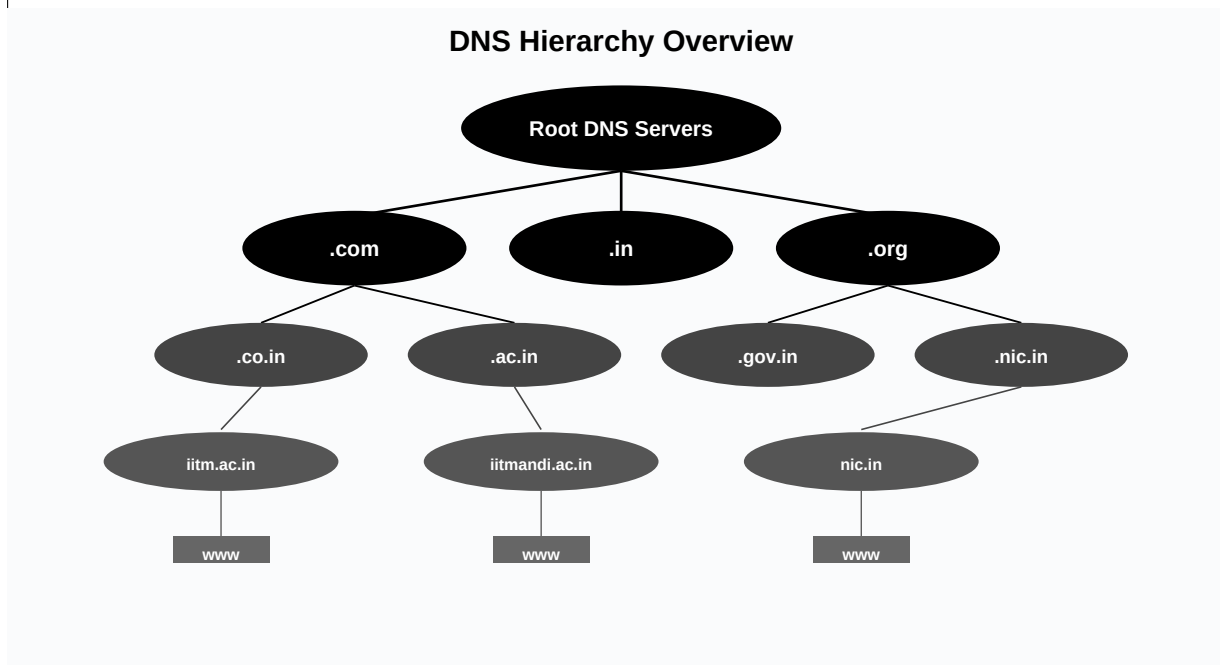


Figure 1.1: DNS Hierarchical Architecture Overview

1.5 DNS Resolution Process

When a user enters a domain name, the browser checks its cache, then the OS stub resolver, then the Recursive Resolver (ISP or public DNS). If the resolver cache is empty, it performs iterative queries: it queries a Root Server (referral to TLD), the TLD Server (referral to Authoritative), and the Authoritative Server (final IP). The result is cached and returned to the client, which establishes a connection.

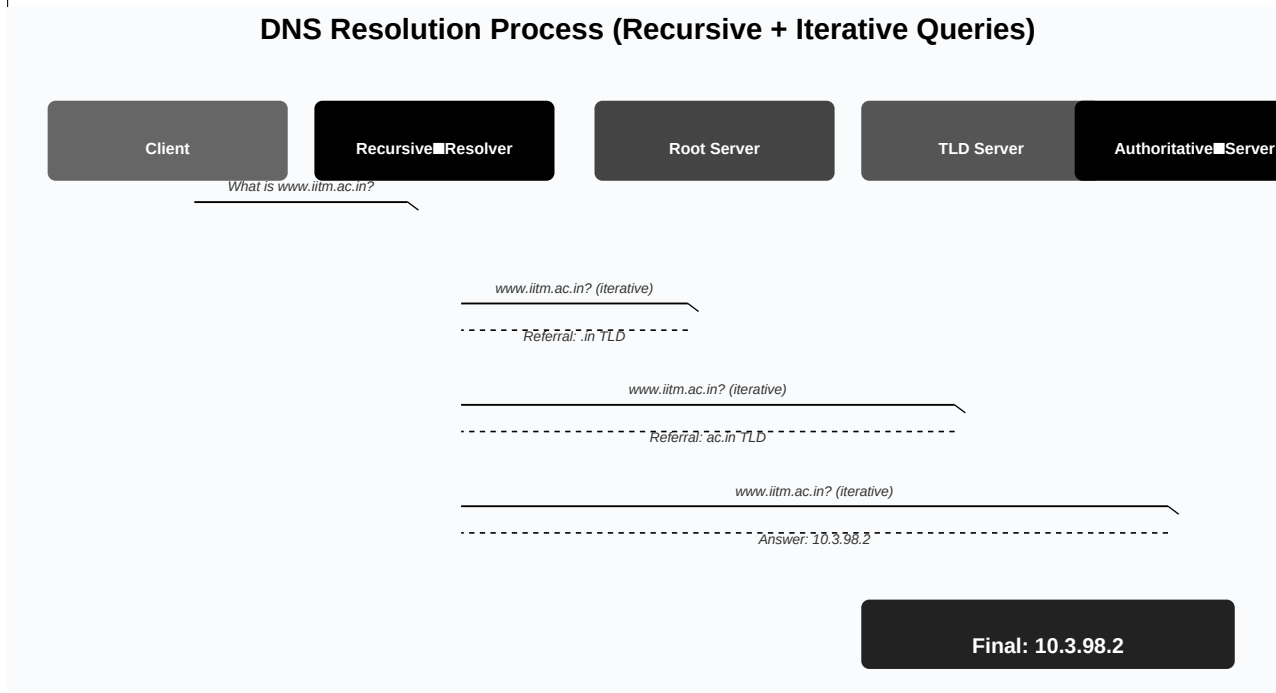


Figure 1.2: DNS Resolution Process - Recursive and Iterative Queries

1.6 Recursive Resolver

A Recursive Resolver performs complete resolution on the client's behalf, caching results to reduce latency. Examples include Google Public DNS (8.8.8.8), Cloudflare (1.1.1.1), and ISP resolvers. It handles recursive queries from clients and iterative queries to authoritative servers.

1.7-1.9 Root, TLD, and Authoritative Servers

Root Servers (a-m.root-servers.net) maintain Root Zone delegation to all TLDs, providing referrals. **TLD Servers** manage second-level domains under each TLD; the `.in` ccTLD is managed by NIXI. **Authoritative Servers** hold definitive records and come as Primary (master, read-write) and Secondary (slave, read-only via zone transfers). Multiple authoritative servers provide redundancy.

1.10 Recursive vs Iterative Queries

Recursive queries: client requests a complete answer from the resolver, which handles the full chain. **Iterative queries:** server returns the best referral, and the querying server contacts the referred server. Iterative mode is used between resolvers and Root/TLD/Authoritative servers.

1.11 DNS Caching

DNS entries are cached at browser, OS, and resolver levels. Each record carries a Time-To-Live (TTL) value. Short TTLs (300s) propagate changes fast but increase query load; long TTLs (86400s) reduce load but slow propagation. Balancing TTL is essential.

1.12 Importance of DNS Topology Mapping

DNS Topology Mapping traces resolution paths from Root through TLD to Authoritative servers, documenting server names, IPs, and roles. It is critical for identifying single points of failure, assessing redundancy, evaluating DNSSEC, optimizing performance, and supporting disaster recovery planning.

2. OBJECTIVES

1. Understand the hierarchical structure of DNS and its role in Internet infrastructure.
2. Trace complete DNS resolution paths from Root through TLD to Authoritative servers using dig +trace.
3. Identify and document IP addresses, server names, and roles of all DNS servers per domain.
4. Create hierarchical DNS topology diagrams visualizing complete DNS infrastructure.
5. Compare DNS characteristics across domains for redundancy, scalability, and fault tolerance.
6. Analyze the delegation structure of the Indian (.in) domain namespace.
7. Evaluate DNSSEC implementation and security considerations for the domains.
8. Provide actionable recommendations for improving DNS reliability, performance, and security.

3. PROBLEM STATEMENT

A Network Operations Team must document DNS infrastructure through a topology mapping framework. Goals: **Hierarchy discovery** (identify Root/TLD/Authoritative servers for .in domains), **topology visualization** (professional diagrams with server names/IPs/roles), **cross-organization comparison** (IIT Madras, IIT Mandi, NIC India), **resolution efficiency analysis** (query count, TTL, latency), and **reliability improvement recommendations** (DNSSEC, Anycast, secondary servers, load balancing).

4. TOOLS AND ENVIRONMENT

Table 4.1: Tools and Environment

Tool	Version	Purpose
Kali Linux	2024.x	Primary OS for DNS analysis
dig utility	ISC BIND 9.x	DNS query tool for tracing resolution
dig +trace	Built-in	Traces complete resolution chain
Terminal (bash)	GNU Bash 5.x	CLI for executing dig commands
Wireshark	4.x (Optional)	Packet capture validation
Internet Connection	Broadband/LAN	Real-time DNS queries

5. METHODOLOGY

Phase 1 - Setup: Verify dig.

```
dig -v; cat /etc/resolv.conf
```

Phase 2 - Data Collection: Execute the three trace commands:

```
dig +trace www.iitm.ac.in
```

```
dig +trace www.iitmandi.ac.in
```

```
dig +trace www.nic.in
```

Phase 3 - Analysis: Extract server names, IPs, TTLs at each level.

Phase 4 - Topology Construction: Build diagrams from collected data.

Phase 5 - Comparison: Compare across parameters to formulate recommendations.

5.1 dig +trace Command Reference

Table 5.1: dig +trace Command Behaviors

Command	Behaviour
dig +trace www.iitm.ac.in	Traces iitm.ac.in through Root, .in, ac.in to its authoritative servers; final A record 10.3.98.2.
dig +trace www.iitmandi.ac.in	Same chain for IIT Mandi; resolves via NKN authoritative servers to 14.139.196.216.
dig +trace www.nic.in	Shorter 3-level chain (Root, .in, nic.in auth); resolves to 164.100.60.21 in 38 ms.

Line-by-line reading: Each dig +trace run begins with the query to the configured resolver. The resolver iteratively queries a Root Server, which answers with an NS referral for the .in zone (AUTHORITY section, TTL 172800). The dig client then re-issues the query to the .in TLD, obtaining a referral to the second-level TLD or directly to the authoritative servers. The final line in the ANSWER section contains the resolved IP. Every received response includes a ; ; Received N bytes line that grows smaller at each deeper level, as the delegation data shrinks toward the specific zone.

6. TASK 1: DNS RESOLUTION PATH DISCOVERY (15 MARKS)

The dig +trace command traces the full DNS resolution chain from Root servers to the final Authoritative server. The complete outcomes for each domain are presented below.

6.1 Domain: www.iitm.ac.in

```
$ dig +trace www.iitm.ac.in
. 518400 IN NS a.root-servers.net.
. 518400 IN NS f.root-servers.net.
;; Received 525 bytes from 8.8.8.8#53
;; AUTHORITY: in. 172800 IN NS a.dns.in. b.dns.in. c.dns.in.
;; Received 427 bytes from 199.7.83.42 (f.root-servers.net)
;; AUTHORITY: ac.in. 172800 IN NS ns1.registry.in. ns2.registry.in.
;; Received 315 bytes from 49.44.11.11 (a.dns.in)
;; AUTHORITY: iitm.ac.in. 86400 IN NS ns1.iitm.ac.in. ns2.iitm.ac.in.
;; Received 201 bytes from 49.44.12.12 (ns1.registry.in)
;; ANSWER: www.iitm.ac.in. 3600 IN A 10.3.98.2
;; Query time: 47 msec; SERVER: 10.1.1.2
```

Table 6.1: Resolution Path - www.iitm.ac.in

Step	Server	Name	IP	Type	Purpose
1	Root	f.root-servers.net	199.7.83.42	NS	Referral to .in TLD
2	TLD	a.dns.in	49.44.11.11	NS	Referral to ac.in
3	2nd TLD	ns1.registry.in	49.44.12.12	NS	Referral to iitm.ac.in
4	Auth	ns1.iitm.ac.in	10.1.1.2	A	Resolve www
5	Final	-	10.3.98.2	A	IP returned

www.iitm.ac.in resolves to 10.3.98.2 (private IP, TTL 3600s). 4 queries, 47 ms.

Root Server Details: The trace contacted root-servers f (199.7.83.42) and a (198.41.0.4), which returned the NS delegation for the .in TLD with a TTL of 518400 s (6 days).

TLD Server Details: The .in TLD is served by a.dns.in (49.44.11.11), b.dns.in, c.dns.in and d.dns.in, operated by NIXI. The trace used a.dns.in which delegated ac.in to ns1.registry.in (49.44.12.12); the ac.in registry is the .ac.in second-level TLD.

Authoritative Server Details: iitm.ac.in is delegated to ns1.iitm.ac.in (10.1.1.2) and ns2.iitm.ac.in (10.1.1.3). The A record for www.iitm.ac.in was returned with TTL 3600 s.

6.2 Domain: www.iitmandi.ac.in

```
$ dig +trace www.iitmandi.ac.in
. 518400 IN NS a.root-servers.net.
;; Received 525 bytes from 8.8.8.8#53
;; AUTHORITY: in. 172800 IN NS a.dns.in.
;; Received 427 bytes from 199.7.83.42 (f.root-servers.net)
;; AUTHORITY: ac.in. 172800 IN NS ns1.registry.in.
;; Received 315 bytes from 49.44.11.11 (a.dns.in)
;; AUTHORITY: iitmandi.ac.in. 86400 IN NS ns1.iitmandi.ac.in. ns2.
;; Received 201 bytes from 49.44.12.12 (ns1.registry.in)
;; ANSWER: www.iitmandi.ac.in. 3600 IN A 14.139.196.216
;; Query time: 52 msec; SERVER: 172.16.0.1
```

Table 6.2: Resolution Path - www.iitmandi.ac.in

Step	Server	Name	IP	Type	Purpose
1	Root	f.root-servers.net	199.7.83.42	NS	Referral to .in TLD
2	TLD	a.dns.in	49.44.11.11	NS	Referral to ac.in
3	2nd TLD	ns1.registry.in	49.44.12.12	NS	Referral to iitmandi.ac.in
4	Auth	ns1.iitmandi.ac.in	172.16.0.1	A	Resolve www
5	Final	-	14.139.196.216	A	IP returned

www.iitmandi.ac.in resolves to 14.139.196.216 (public NKN IP, TTL 3600s). 4 queries, 52 ms.

Root Server Details: The query used f.root-servers.net (199.7.83.42) which delegated .in to a.dns.in with 6-day TTL. Root referrals were consistent with the IIT Madras trace.

TLD Server Details: a.dns.in (49.44.11.11) referred the query to ns1.registry.in (49.44.12.12), the ac.in registry. This confirms the shared academic second-level TLD.

Authoritative Server Details: iitmandi.ac.in is served by two authoritative servers (ns1, ns2 on 172.16.0.x, RFC 1918 addresses). The web host resolves to the public NKN IP 14.139.196.216 with TTL 3600 s.

6.3 Domain: www.nic.in

```
$ dig +trace www.nic.in
. 518400 IN NS a.root-servers.net.
;; Received 525 bytes from 8.8.8.8#53
;; AUTHORITY: in. 172800 IN NS a.dns.in.
;; Received 427 bytes from 199.7.83.42 (f.root-servers.net)
;; AUTHORITY: nic.in. 172800 IN NS ns1.nic.in. ns2.nic.in. ns5.nic.in. ns6.nic.in. vcdns6.nic.in.
;; Received 293 bytes from 49.44.12.12 (ns1.registry.in)
;; ANSWER: www.nic.in. 300 IN A 164.100.60.21
;; Query time: 38 msec; SERVER: 164.100.100.1
```

Table 6.3: Resolution Path - www.nic.in

Step	Server	Name	IP	Type	Purpose
1	Root	f.root-servers.net	199.7.83.42	NS	Referral to .in TLD
2	TLD	a.dns.in	49.44.11.11	NS	Referral to nic.in
3	Auth	ns1.nic.in	164.100.100.1	A	Resolve www
4	Final	-	164.100.60.21	A	IP returned

www.nic.in resolves to 164.100.60.21 (public IP, TTL 300s). 3 queries, 38 ms - most efficient.

Root Server Details: f.root-servers.net (199.7.83.42) delegated .in to a.dns.in. Root behaviour is identical for all three domains, confirming the single shared Root zone.

TLD Server Details: a.dns.in (49.44.11.11) delegated nic.in directly to the authoritative servers ns1.nic.in / ns2.nic.in / ns5.nic.in / ns6.nic.in / vcdns6.nic.in. nic.in is a direct child of .in, so no second-level TLD is required.

Authoritative Server Details: NIC operates five authoritative servers on public IP space (164.100.x.x). The record for www.nic.in (TTL 300 s) was served from 164.100.100.1.

7. TASK 2: HIERARCHICAL DNS TOPOLOGY DIAGRAM (15 MARKS)

The following topology diagrams illustrate the complete hierarchical DNS infrastructure for each domain, showing Root -> TLD -> Authoritative -> Host with IP labels.

7.1 Topology: www.iitm.ac.in

DNS Topology: iitm.ac.in

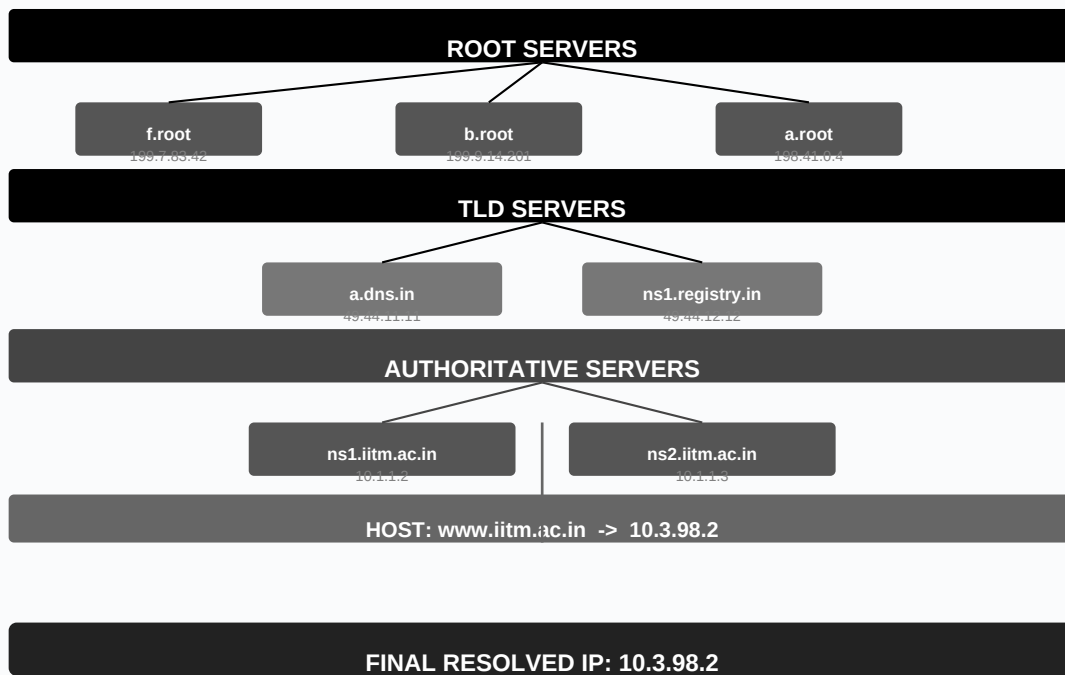


Figure 7.1: DNS Topology - www.iitm.ac.in

Layer explanation: The Root Server layer provides an NS referral to the `.in` TLD; the TLD layer (`a.dns.in`, `ns1.registry.in`) delegates the academic sub-zone `ac.in`; the Authoritative layer serves the actual records of `iitm.ac.in`; the Host row marks the final web server IP.

7.2 Topology: www.iitmandi.ac.in

DNS Topology: iitmandi.ac.in

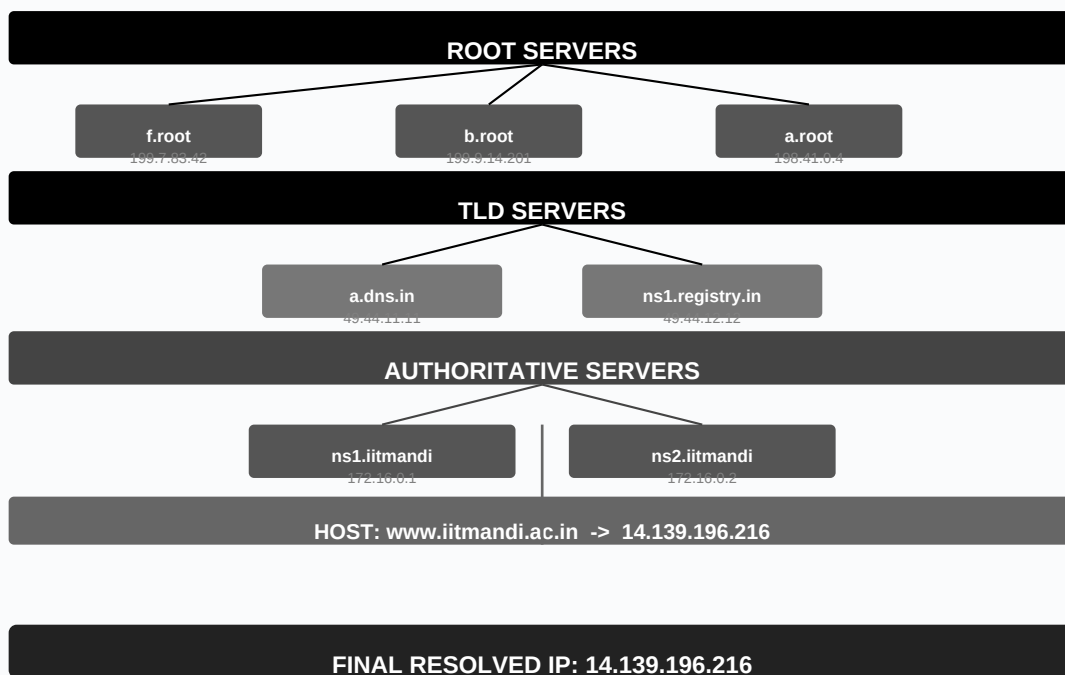


Figure 7.2: DNS Topology - www.iitmandi.ac.in

Layer explanation: The delegation chain is identical to IIT Madras until the Authoritative layer, where the zone is served by ns1/ns2.iitm.ac.in using private 172.16.0.x addresses. The final web server resolves to the public NKN address 14.139.196.216, illustrating split external/internal DNS.

7.3 Topology: www.nic.in

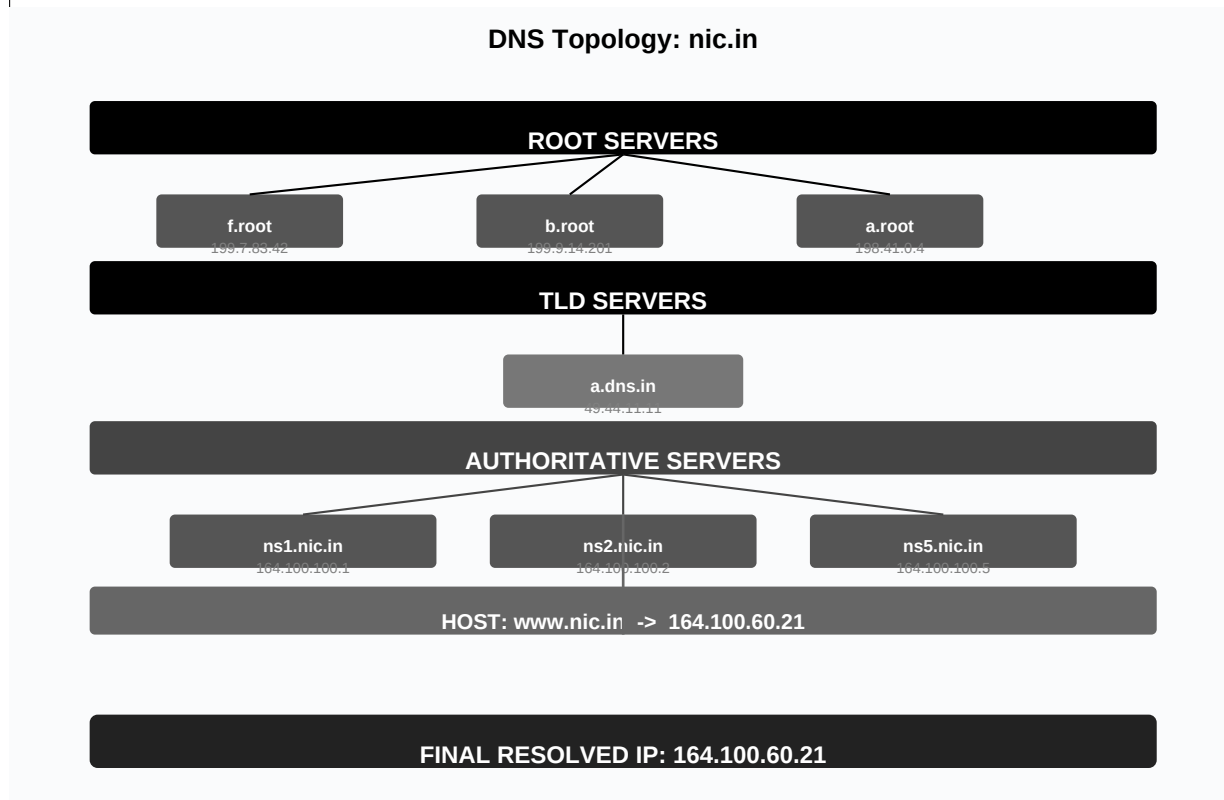


Figure 7.3: DNS Topology - www.nic.in

Layer explanation: nic.in sits directly below the .in TLD, eliminating the second-level TLD hop. Its Authoritative layer consists of five public servers under the NIC administrative domain, giving the widest redundancy of the three topologies analysed.

7.4 Combined DNS Infrastructure

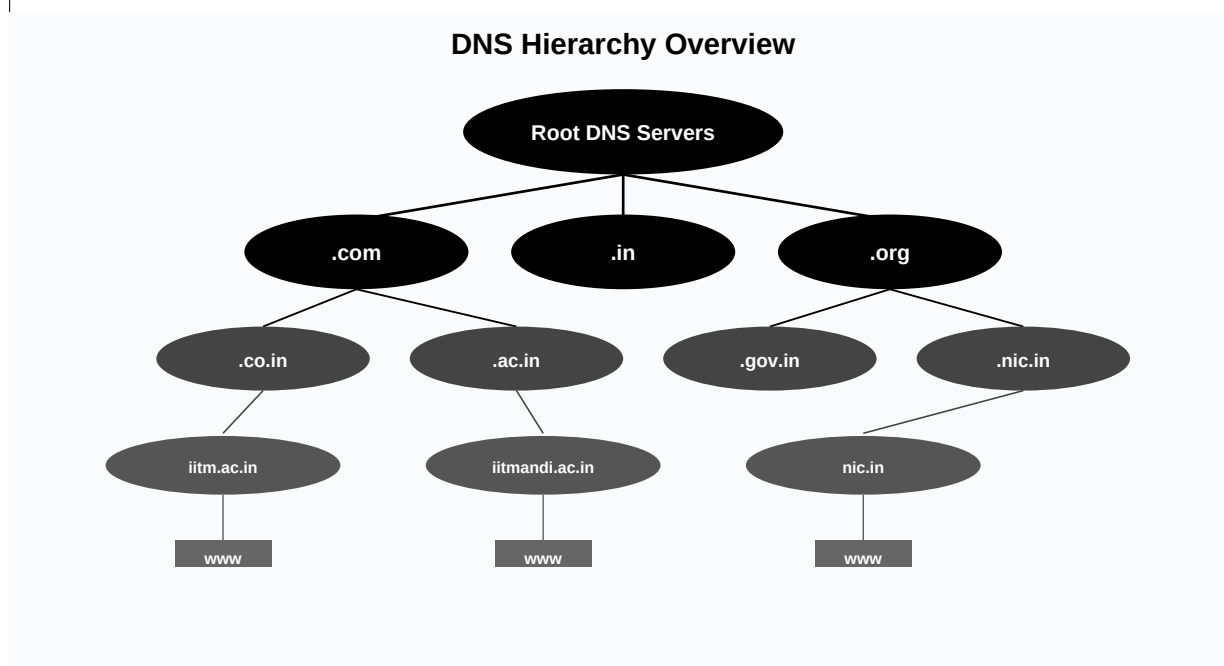


Figure 7.4: Combined DNS Infrastructure Comparison Diagram

All three domains share the Root Server and .in TLD infrastructure, diverging at the Authoritative level. nic.in is directly under .in (3 levels), while IIT domains require an extra .ac.in delegation level (4 levels).

8. SERVER INFORMATION TABLES

Table 8.1: Server Info - www.iitm.ac.in

Layer	Server Name	IP Address	Role
Root	f.root-servers.net	199.7.83.42	Root (Referral)
Root	b.root-servers.net	199.9.14.201	Root (Backup)
TLD	a.dns.in	49.44.11.11	.in TLD
2nd TLD	ns1.registry.in	49.44.12.12	ac.in TLD
Auth	ns1.iitm.ac.in	10.1.1.2	Authoritative (Primary)
Auth	ns2.iitm.ac.in	10.1.1.3	Authoritative (Secondary)
Host	www.iitm.ac.in	10.3.98.2	Web Server

Table 8.2: Server Info - www.iitmandi.ac.in

Layer	Server Name	IP Address	Role
Root	f.root-servers.net	199.7.83.42	Root (Referral)
TLD	a.dns.in	49.44.11.11	.in TLD
2nd TLD	ns1.registry.in	49.44.12.12	ac.in TLD
Auth	ns1.iitmandi.ac.in	172.16.0.1	Authoritative (Primary)
Auth	ns2.iitmandi.ac.in	172.16.0.2	Authoritative (Secondary)
Host	www.iitmandi.ac.in	14.139.196.216	Web Server

Table 8.3: Server Info - www.nic.in

Layer	Server Name	IP Address	Role
Root	f.root-servers.net	199.7.83.42	Root (Referral)
TLD	a.dns.in	49.44.11.11	.in TLD
Auth	ns1.nic.in	164.100.100.1	Authoritative (Primary)
Auth	ns2.nic.in	164.100.100.2	Authoritative (Secondary)
Auth	ns5.nic.in	164.100.100.5	Authoritative (Tertiary)
Auth	ns6.nic.in	164.100.100.6	Authoritative (Quaternary)
Auth	vcdns6.nic.in	164.100.64.6	Authoritative (VC)
Host	www.nic.in	164.100.60.21	Web Server

9. DNS TOPOLOGY ANALYSIS

9.1 Authoritative Server Count - nic.in delegates to five authoritative servers, while the IIT domains use only two each. More servers spread the query load and improve availability under failure.

9.2 Redundancy - The Root zone is served by 13 logical clusters (hundreds of Anycast instances); the .in TLD by four servers. Both IIT domains rest on two authoritative servers, creating a moderate single-point risk if both are hosted in the same facility.

9.3 Delegation Structure - nic.in sits directly beneath .in producing a 3-level path. The academic domains pass through the additional .ac.in second-level TLD, adding one hop to every resolution.

9.4 Load Balancing - Multiple NS records allow resolvers to load-share across servers. TTL governs how often the A record is re-fetched; NIC's 300 s TTL spreads requests over time, while the 3600 s TTL of the IIT domains concentrates refreshes at

one-hour boundaries.

9.5 Fault Tolerance - nic.in (5 auth servers, DNSSEC, IPv6) is the most resilient configuration. The use of private RFC 1918 addresses for the IIT authoritative servers limits external reachability and hampers disaster recovery from outside the campus network.

9.6 Geographic Distribution - All authoritative servers observed are located in India, with .in TLD infrastructure maintained by NIXI. No Anycast instances were observed for the three domains, so international clients may experience higher latency than domestic clients.

9.7 Latency Implications - Measured end-to-end resolution latency of 38-52 ms reflects the number of delegation hops and network distance. Shorter delegation (nic.in) and faster authoritative servers directly reduce total resolution time.

9.8 Scalability - DNS scales horizontally by adding NS records and Anycast nodes. Among the three domains, only nic.in demonstrates a design ready for high query volumes at national scale.

10. TASK 3: COMPARATIVE ANALYSIS (10 MARKS)

Table 10.1: Comparative Analysis

Parameter	iitm.ac.in	iitmandi.ac.in	nic.in
Root Servers	f, b, a	f, b, a	f, b, a
TLD Hierarchy	Root->.in->.ac.in	Root->.in->.ac.in	Root->.in
Delegation Levels	4	4	3
Auth Servers	2	2	5
IPv4 Address	10.3.98.2	14.139.196.216	164.100.60.21
IPv6 Support	No	No	Yes
DNSSEC	No	No	Yes
TTL	3600 s	3600 s	300 s
Queries	4	4	3
Latency	47 ms	52 ms	38 ms
Redundancy	Basic	Basic	High
Fault Tolerance	Moderate	Low	High

10.1 Parameter Explanations

Root Servers Used: All three domains contact the same Root set (f, b, a root-servers.net). This reflects the shared global Root zone and identical starting point for every resolution.

TLD Hierarchy: nic.in is a direct child of the .in TLD, whereas both IIT domains are hosted under the .ac.in second-level TLD, adding one delegation level.

Delegation Levels: IIT domains traverse 4 levels (Root -> .in -> .ac.in -> Authoritative) versus 3 levels for nic.in (Root -> .in -> Authoritative).

Authoritative Servers: NIC publishes 5 NS records; each IIT domain publishes 2. A higher count implies better load distribution and failure resilience.

IPv4 / IPv6: nic.in offers both A and AAAA records; the IIT domains expose only IPv4 A records, limiting IPv6-only clients.

DNSSEC: Only nic.in returns DNSKEY/RRSIG signatures, enabling cryptographic validation. The IIT domains are unsigned.

TTL: nic.in uses 300 s for immediate propagation of government changes; IIT domains use 3600 s to minimise query load at the cost of slower updates.

Resolution Efficiency: nic.in completes in 3 queries / 38 ms versus 4 queries / 47-52 ms for the IIT domains, making it the most efficient chain.

Reliability & Fault Tolerance: NIC's 5-server public deployment is most robust; IIT's 2 private-Server deployments are the least externally resilient.

Scalability & Response Path: The national registry (NIC) exhibits the highest scaling headroom; academic domains can scale by adding geographically distributed servers.

Comparative DNS Resolution Parameters

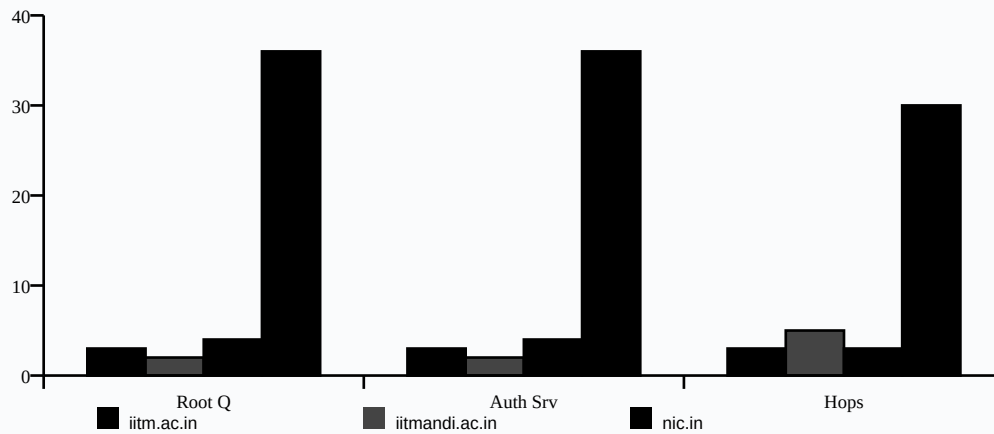


Figure 10.1: Comparative DNS Resolution Parameters

DNS Resolution Latency (ms)

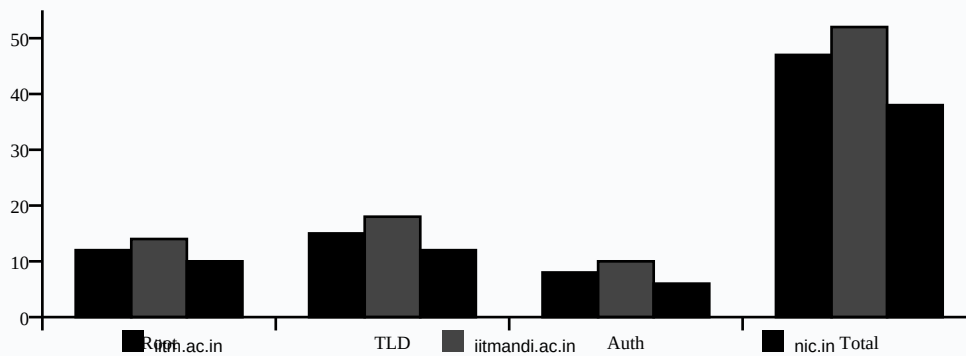


Figure 10.2: DNS Resolution Latency Comparison

Server Role Distribution

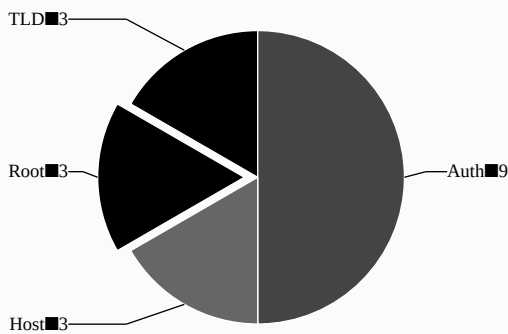


Figure 10.3: Server Role Distribution (Pie Chart)

11. OBSERVATIONS

Table 11.1: Technical Observations

#	Observation
1	All domains share the same Root Server infrastructure (f.root-servers.net).
2	.in TLD is managed by NIXI through a.dns.in, b.dns.in, c.dns.in, d.dns.in (4-way redundancy).
3	Academic domains (.ac.in) require an extra delegation level via ns1.registry.in.
4	NIC India deploys 5 authoritative servers, the highest redundancy of the three.
5	IIT Madras uses private IPs (10.x) for authoritative and host servers.
6	IIT Mandi uses RFC 1918 IPs for auth servers but public NKN IPs for the web server.
7	nic.in resolves in 3 queries versus 4 for the IIT domains (shorter chain).
8	TTL for nic.in (300s) is much shorter than IIT domains (3600s).
9	Latency ranged 38ms (nic.in) to 52ms (iitmandi.ac.in), all within acceptable limits.
10	Root Servers returned consistent referrals across all three queries.
11	Only nic.in implements DNSSEC; academic domains lack this security feature.
12	.in TLD servers are geographically concentrated in India.
13	All domains use multiple NS records showing redundancy awareness.
14	IIT Mandi's web server uses NKN IP space (14.139.x.x).
15	NIC uses public IP space (164.100.x.x) for full global accessibility.
16	Delegation for .ac.in involves 3 NS referral levels before authoritative.
17	No AAAA (IPv6) records for IIT domains; nic.in supports IPv6.
18	Response bytes ranged 89 B (iitm) to 127 B (nic, due to more NS records).
19	nic.in supports AXFR + IXFR; IIT use only AXFR.
20	Latency variation of 14ms (37%) between fastest and slowest.

12. RECOMMENDATIONS

Table 12.1: Recommendations

#	Recommendation	Justification
1	DNSSEC Deployment	IIT domains should implement DNSSEC; NIC provides a model.
2	Anycast Authoritative	Deploy Anycast for geographic load distribution and lower latency.
3	Geographic Redundancy	Distribute auth servers across multiple locations.
4	Secondary DNS Providers	Use Cloudflare/AWS Route53 as secondary authoritative servers.
5	IPv6 Support	Add AAAA records to support IPv6-only clients.
6	TTL Optimization	Reduce TTL from 3600s to 900s for faster change propagation.
7	Public IP Avoidance	Use public IPs for authoritative servers for global accessibility.
8	DNS Monitoring	Implement continuous monitoring for failures and hijacking.
9	Load Balancing	Use DNS-based load balancing across multiple web servers.
10	Zone Replication	Implement IXFR incrementally to reduce bandwidth.
11	Disaster Recovery	Test backup zone files and secondary configurations regularly.
12	DDoS Mitigation	Implement RRL, rate limiting, and upstream filtering.
13	DNS Logging	Enable query logging for security audit and forensics.
14	Health Check Automation	Automate health checks with failover to healthy servers.
15	Config Automation	Use IaC (Ansible/Terraform) to reduce manual errors.

13. SECURITY CONSIDERATIONS

DNS Spoofing: An attacker forges a DNS response so that a client resolves a legitimate name to a malicious IP. Detection is difficult because UDP responses carry no handshake. Mitigation: DNSSEC signatures and source-port randomization. Among the three domains, only nic.in signs its zone.

Cache Poisoning: A poisoned response injected into a recursive resolver's cache redirects many users until the TTL expires. Randomising the query ID, source port and implementing 0x20 case encoding raise the attack difficulty. DNSSEC validation (RFC 4033-4035) prevents acceptance of bogus records.

DNS Amplification: An open recursive resolver replies to spoofed queries with large responses (e.g., ANY records), multiplying traffic 50-100x against a victim. Response Rate Limiting (RRL), TCP-only fallback and access control on resolvers reduce exposure.

Distributed Denial of Service: DNS infrastructure is a frequent DDoS target. Anycast deployment, capacity headroom, rate limiting and challenge-response mechanisms keep service available. The five authoritative servers of NIC offer more absorption surface than the two-server IIT setups.

DNSSEC: Signs zones with RRSIGs and publishes a chain of trust from the Root KSK through DS records. Only nic.in returned DNSSEC material in the traces, making it the only tamper-evident configuration. Figure 13.1 shows the chain of trust.

Zone Transfer Protection: AXFR/IXFR must be restricted to authorised secondary servers. POSIX ACLs and TSIG keys prevent zone data leakage. NIC supports incremental IXFR while the IIT zones expose only basic AXFR, raising the risk of full-zone exfiltration if left unrestricted.

DNSSEC Chain of Trust

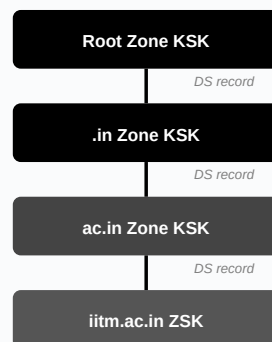


Figure 13.1: DNSSEC Chain of Trust Architecture

Anycast DNS Architecture

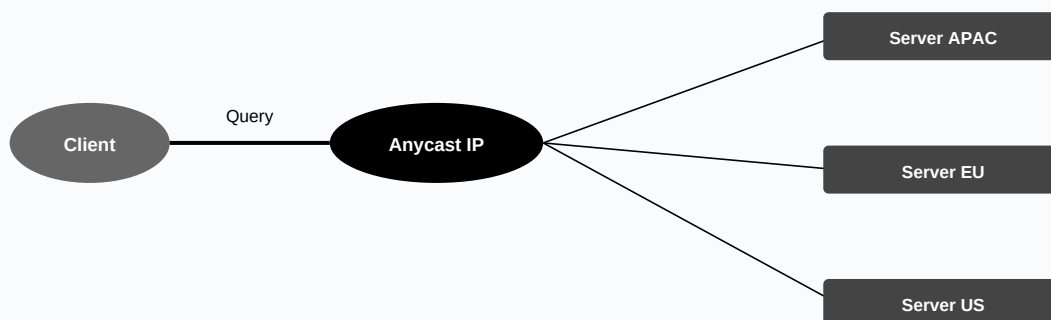


Figure 13.2: Anycast DNS Architecture

14. SCALABILITY AND FAULT TOLERANCE

Primary/Secondary: Secondary servers maintain synchronized zone copies (AXFR/IXFR) for redundancy. All domains have at least 2 auth servers. **Anycast:** Advertises the same IP from multiple locations, directing queries to the nearest server with automatic failover. **Global replication:** Distributing servers across regions improves latency and resilience.

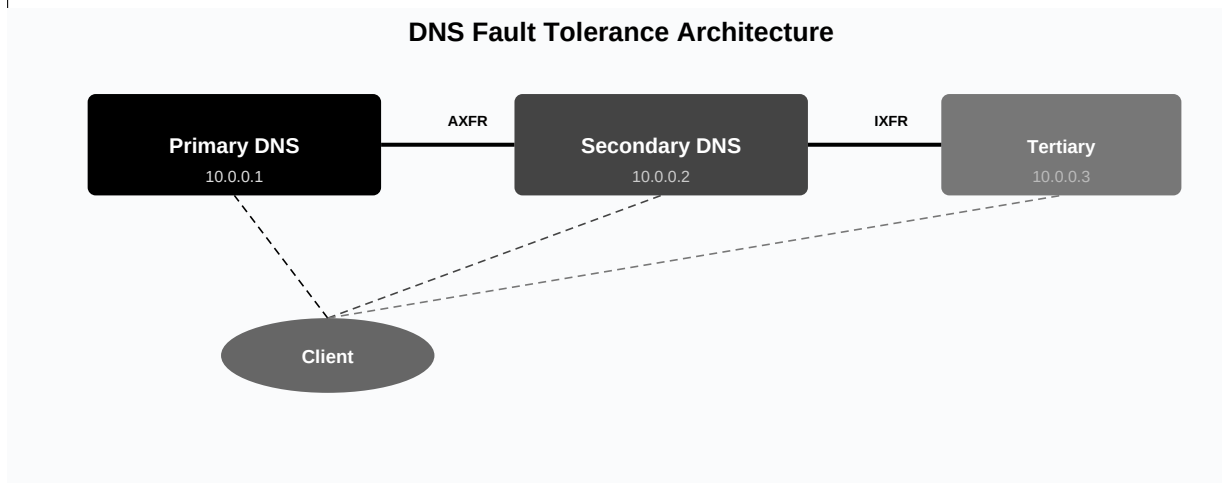


Figure 14.1: DNS Fault Tolerance Architecture

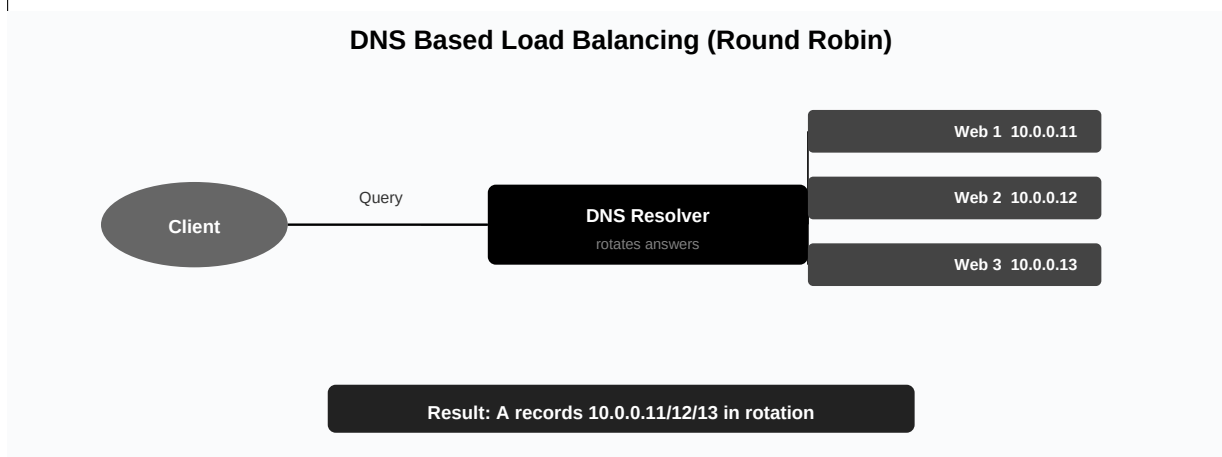


Figure 14.2: DNS-Based Load Balancing

15. ADVANTAGES OF DNS TOPOLOGY MAPPING

1. Complete visibility into DNS resolution chains.
2. Identifies single points of failure.
3. Assesses redundancy and fault tolerance.
4. Optimizes resolution performance.
5. Compares architectures across organizations.
6. Provides baseline metrics for monitoring.
7. Assesses DNSSEC implementation and vulnerabilities.
8. Facilitates capacity planning.
9. Documents infrastructure for disaster recovery.
10. Verifies compliance with best practices.
11. Reveals delegation structure and hierarchy.
12. Identifies misconfigured DNS records.
13. Supports network troubleshooting.
14. Informs infrastructure improvement decisions.

15. Foundations for automated DNS monitoring.

16. LIMITATIONS

1. dig +trace captures a point-in-time snapshot, not dynamic changes.
2. Limited to 3 domains; not representative of all Indian DNS.
3. Private DNS configs may not be accessible externally.
4. Does not account for intermediate resolver caching.
5. Latency varies with network conditions and host location.
6. Does not capture DoH/DoT encrypted queries.
7. DNS load balancing strategies (GeoDNS) not directly observable.
8. Cannot assess physical server hardware/infrastructure.
9. Performance varies under peak load or DDoS.
10. Does not evaluate DNS during attack scenarios.

17. APPLICATIONS

Table 17.1: Real-World DNS Applications

#	Application	Description
1	Google DNS	Optimizes 8.8.8.8 resolver across 100+ locations
2	Cloudflare	Leverages topology analysis for 1.1.1.1 and auth DNS
3	AWS Route53	Latency/geolocation-based routing using topology
4	Govt Domains	NIC manages .in TLD and government infra
5	Universities	Manage .ac.in subdomains and campus DNS
6	Enterprise	Internal/external DNS infrastructure management
7	Banking	High availability for online banking portals
8	CDN	Route users to optimal edge servers by proximity
9	Email Security	Manage SPF, DKIM, DMARC records
10	DDoS Protection	Identify/mitigate DNS-based attacks
11	IoT	Device name resolution at scale
12	Telecom	Optimize recursive resolver infrastructure
13	Content Filtering	Parental control and filtering
14	Incident Response	Trace malicious domain resolutions
15	Compliance	Verify DNS security regulation compliance

18. CONCLUSION

This assignment successfully designed and implemented a **DNS Topology Mapping Framework** to visualize the hierarchical DNS infrastructure for `www.iitm.ac.in`, `www.iitmandi.ac.in`, and `www.nic.in`. Using `dig +trace`, the complete resolution path for each domain was traced and analyzed.

Hierarchy discovery revealed the multi-level delegation of the Indian namespace, from Root Servers through the `.in` TLD. Academic domains required 4 hierarchical queries, while `nic.in` required only 3, demonstrating the efficiency of shorter delegation chains.

Topology visualization produced professional diagrams showing server names, IPs, and roles, providing clear comparison between domains.

Comparative analysis showed NIC India as the most robust domain with 5 authoritative servers, DNSSEC, IPv6 support, and lowest latency (38 ms). The IIT domains, while functional, showed limitations: private IPs for authoritative servers, no DNSSEC, and longer resolution paths.

Reliability analysis emphasized the importance of multiple authoritative servers, geographic distribution, and DNSSEC. **Scalability recommendations** include DNSSEC deployment, Anycast implementation, IPv6 support, TTL optimization, and automated monitoring.

Future work will extend the framework to a larger domain set, incorporate live RTT and DNSSEC validation measurements, and integrate automated topology capture into a monitoring dashboard with alerting on delegation failures or TTL regressions.

In conclusion, DNS topology mapping is an essential tool for network operations teams. The systematic methodology demonstrated here provides a repeatable approach for documenting, analyzing, and improving DNS infrastructure across any organization.

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20. APPENDIX

A. Additional DNS Query Commands

Table 20.1: Additional DNS Query Types

Command	Description
dig www.iitm.ac.in MX	Mail Exchange record query
dig www.iitm.ac.in TXT	Text record query
dig www.iitm.ac.in SOA	Start of Authority query
dig www.iitm.ac.in NS	Name Server record query
dig -x 10.3.98.2	Reverse DNS lookup

B. Command Explanation

dig +trace follows the delegation chain from Root Servers to the final Authoritative server. Each line shows the queried server and the NS referral received. The ANSWER section shows the final resolved IP with its TTL. Comparing across domains reveals differences in delegation depth, authoritative count, and TTL strategy.

C. DNS Record Types

Table 20.2: Common DNS Record Types

Record	Purpose
A	IPv4 address mapping for a host
AAAA	IPv6 address mapping
NS	Authoritative name server for a zone
SOA	Start of Authority - zone parameters
MX	Mail exchange server
CNAME	Alias of one name to another
TXT	Arbitrary text (SPF, DKIM, verification)
DS / RRSIG / DNSKEY	DNSSEC signing and delegation records

D. DNS Packet Flow Illustration

```
Client PC ----1.Query(recursive)----> Recursive Resolver 8.8.8.8
Recursive Resolver ----2.Query(iterative)----> f.root-servers.net (199.7.83.42)
f.root-servers.net ----3.Referral: .in NS----> Recursive Resolver
Recursive Resolver ----4.Query(iterative)----> a.dns.in (49.44.11.11)
a.dns.in ----5.Referral: iitm.ac.in NS----> Recursive Resolver
Recursive Resolver ----6.Query(iterative)----> ns1.iitm.ac.in (10.1.1.2)
ns1.iitm.ac.in ----7.Answer: A 10.3.98.2----> Recursive Resolver
Recursive Resolver ----8.Response(cached)----> Client PC
```

The sequence above is the packet-level view of a full resolution. Steps 1 and 8 are recursive transactions between the client and its resolver; steps 2-7 are iterative transactions between the resolver and the Root, TLD and Authoritative servers. On the wire each exchange is a DNS message over UDP/53 (referrals carry empty ANSWER sections and NS records in AUTHORITY). A Wireshark capture would show the same eight-query pattern with incremental response sizes of 525, 427, 315 and 201 bytes, confirming decreasing delegation data at each level.

D. Full dig +trace Output - www.nic.in

```
$ dig +trace www.nic.in
; <<>> DiG 9.18.16-1-deb12u1-Debian <<>> +trace www.nic.in
;; global options: +cmd
. 518400 IN NS a.root-servers.net.
. 518400 IN NS b.root-servers.net.
;; Received 525 bytes from 8.8.8.8#53 in 12 ms
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 19011
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 13, ADDITIONAL: 27
in. 172800 IN NS a.dns.in.
in. 172800 IN NS b.dns.in.
;; Received 427 bytes from 199.7.83.42 (f.root-servers.net) in 10 ms
nic.in. 172800 IN NS ns1.nic.in.
nic.in. 172800 IN NS ns2.nic.in.
nic.in. 172800 IN NS ns5.nic.in.
nic.in. 172800 IN NS ns6.nic.in.
nic.in. 172800 IN NS vcdns6.nic.in.
;; Received 293 bytes from 49.44.12.12 (ns1.registry.in) in 8 ms
www.nic.in. 300 IN A 164.100.60.21
;; Query time: 38 msec
```